

Name:	ID:	Section:
-------	-----	----------

Question 1 [C02] [10 Points]

Design the Anime class in such a way that the following code provides the expected output. **[An Anime can have up to 10 episodes.]**

Driver Class	Output
<pre> public class AnimeTester { public static void main(String[] args) { Anime a1 = new Anime("Jujutsu Kaisen 0"); Anime a2 = new Anime("Jujutsu Kaisen", "Shounen"); a1.display(); a2.display(); System.out.println("====="); a1.addEpisode("The Night Parade", 105, true); a1.addEpisode("Hidden Inventory", 45, false); a2.addEpisode("Transformation", 24, true); a2.addEpisode("Hollow Purple"); System.out.println("====="); a1.display(); a2.display(); System.out.println("====="); a2.setPrevious(a1); a1.rateEpisode("The Night Parade", 7.0); a2.rateEpisode("Transformation", 8.5); a2.rateEpisode("Hollow Purple", 9.8); a1.display(); a2.display(); System.out.println("====="); a2.canonDuration(); a2.topRated(); } } </pre>	<pre> Anime: Jujutsu Kaisen 0 Genre: Unknown Previous: None Anime: Jujutsu Kaisen Genre: Shounen Previous: None ===== Added: The Night Parade Duration: 105 min Canon: Yes Added: Hidden Inventory Duration: 45 min Canon: No Added: Transformation Duration: 24 min Canon: Yes Added: Hollow Purple Duration: 18 min Canon: Yes ===== Anime: Jujutsu Kaisen 0 Genre: Unknown Previous: None The Night Parade Rating: 0.0/10 Hidden Inventory Rating: 0.0/10 Anime: Jujutsu Kaisen Genre: Shounen Previous: None Transformation Rating: 0.0/10 Hollow Purple Rating: 0.0/10 ===== Anime: Jujutsu Kaisen 0 Genre: Unknown Previous: None The Night Parade Rating: 7.0/10 Hidden Inventory Rating: 0.0/10 Anime: Jujutsu Kaisen Genre: Shounen Previous: Jujutsu Kaisen 0 Transformation Rating: 8.5/10 Hollow Purple Rating: 9.8/10 ===== Total Canon Duration of Jujutsu Kaisen: 42 min Top Rated Episode of Jujutsu Kaisen: Hollow Purple (9.8/10) </pre>

Name:	ID:	Section:
-------	-----	----------

Question 1 [C02] [10 Points]

Design the Anime class in such a way that the following code provides the expected output. [An Anime can have up to 12 episodes.]

Driver Class	Output
<pre> public class AnimeTester { public static void main(String[] args) { Anime a1 = new Anime("Tokyo Ghoul"); Anime a2 = new Anime("Tokyo Ghoul:re", "Seinen"); a1.display(); a2.display(); System.out.println("====="); a1.addEpisode("Tragedy", 48, true); a1.addEpisode("Ghoul", 26, false); a2.addEpisode("Haise Sasaki", 22, true); a2.addEpisode("One Eyed King"); System.out.println("====="); a1.display(); a2.display(); System.out.println("====="); a2.setPrevious(a1); a1.rateEpisode("Tragedy", 8.7); a2.rateEpisode("Haise Sasaki", 9.2); a2.rateEpisode("One Eyed King", 7.5); a1.display(); a2.display(); System.out.println("====="); a2.canonDuration(); a2.topRated(); } } </pre>	<pre> Anime: Tokyo Ghoul Genre: Unknown Previous: None Anime: Tokyo Ghoul:re Genre: Seinen Previous: None ===== Added: Tragedy Duration: 48 min Canon: Yes Added: Ghoul Duration: 26 min Canon: No Added: Haise Sasaki Duration: 22 min Canon: Yes Added: One Eyed King Duration: 20 min Canon: No ===== Anime: Tokyo Ghoul Genre: Unknown Previous: None Tragedy Rating: 0.0/10 Ghoul Rating: 0.0/10 Anime: Tokyo Ghoul:re Genre: Seinen Previous: None Haise Sasaki Rating: 0.0/10 One Eyed King Rating: 0.0/10 ===== Anime: Tokyo Ghoul Genre: Unknown Previous: None Tragedy Rating: 8.7/10 Ghoul Rating: 0.0/10 Anime: Tokyo Ghoul:re Genre: Seinen Previous: Tokyo Ghoul Haise Sasaki Rating: 9.2/10 One Eyed King Rating: 7.5/10 ===== Total Canon Duration of Tokyo Ghoul:re: 22 min Top Rated Episode of Tokyo Ghoul:re: Haise Sasaki (9.2/10) </pre>

SET A

```
public class Anime {

    String title;
    String genre = "Unknown";
    Anime previous;

    String[] episodeNames = new String[20];
    int[] durations = new int[20];
    boolean[] canon = new boolean[20];
    double[] ratings = new double[20];
    int count = 0;

    public Anime(String title) {
        this.title = title;
    }

    public Anime(String title, String genre) {
        this.title = title;
        this.genre = genre;
    }

    public void addEpisode(String name) {
        addEpisode(name, 18, true);
    }

    public void addEpisode(String name, int duration, boolean isCanon) {
        episodeNames[count] = name;
        durations[count] = duration;
        canon[count] = isCanon;
        ratings[count] = 0.0;

        System.out.println("Added: " + name + " | Duration: " +
            duration + " min | Canon: " +
            (isCanon ? "Yes" : "No"));

        count++;
    }

    public void setPrevious(Anime a) {
        previous = a;
    }

    public void rateEpisode(String name, double rating) {
        for (int i = 0; i < count; i++) {
            if (episodeNames[i].equals(name)) {
                ratings[i] = rating;
                break;
            }
        }
    }
}
```

```

    }
}

public void display() {
    System.out.print("Anime: " + title +
        " | Genre: " + genre +
        " | Previous: ");

    if (previous == null)
        System.out.println("None");
    else
        System.out.println(previous.title);
    for (int i = 0; i < count; i++) {
        System.out.println(" [" + (i + 1) + "] " +
            episodeNames[i] +
            " | Rating: " + ratings[i] + "/10");
    }
}

public void canonDuration() {
    int total = 0;

    for (int i = 0; i < count; i++) {
        if (canon[i]) {
            total += durations[i];
        }
    }
    System.out.println("Total Canon Duration of " + title +
        ": " + total + " min");
}

public void topRated() {
    int max = 0;
    for (int i = 1; i < count; i++) {
        if (ratings[i] > ratings[max]) {
            max = i;
        }
    }
    System.out.println("Top Rated Episode of " + title + ": " +
        episodeNames[max] + " (" + ratings[max] + "/10)");
}
}

```

SET B

```
public class Anime {

    String title;
    String genre = "Unknown";
    Anime previous;

    String[] episodeNames = new String[20];
    int[] durations = new int[20];
    boolean[] canon = new boolean[20];
    double[] ratings = new double[20];
    int count = 0;

    public Anime(String title) {
        this.title = title;
    }

    public Anime(String title, String genre) {
        this.title = title;
        this.genre = genre;
    }

    public void addEpisode(String name) {
        addEpisode(name, 20, false);
    }

    public void addEpisode(String name, int duration, boolean isCanon) {
        episodeNames[count] = name;
        durations[count] = duration;
        canon[count] = isCanon;
        ratings[count] = 0.0;
        System.out.println("Added: " + name +
            " | Duration: " + duration +
            " min | Canon: " + (isCanon ? "Yes" : "No"));
        count++;
    }

    public void setPrevious(Anime a) {
        previous = a;
    }

    public void rateEpisode(String name, double rating) {
        for (int i = 0; i < count; i++) {
            if (episodeNames[i].equals(name)) {
                ratings[i] = rating;
                return;
            }
        }
    }
}
```

```

public void display() {
    System.out.print("Anime: " + title +
        " | Genre: " + genre +
        " | Previous: ");

    if (previous == null)
        System.out.println("None");
    else
        System.out.println(previous.title);
    for (int i = 0; i < count; i++) {
        System.out.println(" [" + (i + 1) + "] " +
            episodeNames[i] +
            " | Rating: " + ratings[i] + "/10");
    }
}

public void canonDuration() {
    int total = 0;
    for (int i = 0; i < count; i++) {
        if (canon[i]) {
            total += durations[i];
        }
    }
    System.out.println("Total Canon Duration of " + title + ": " + total + " min");
}

public void topRated() {
    int max = 0;
    for (int i = 1; i < count; i++) {
        if (ratings[i] > ratings[max]) {
            max = i;
        }
    }
    System.out.println("Top Rated Episode of " + title + ": " +
        episodeNames[max] + " (" + ratings[max] + "/10)");
}
}

```

RUBRICS:

Correct instance variables	- 1 mark
Constructor Overloading	- 1 mark
addEpisode() method overloading	- 0.5 mark for separate method

	- 1.5 marks for updating variables & printing
display()	- 1.5 marks correct printing
setPrevious()	- Object reference 1 mark
rateEpisode()	- Update correct anime by name 1 mark
canonDuration()	- Calculate total canon duration iteratively 1.5 marks
topRated()	- Correct maximum rated episode 1 mark